

# Estuardo Lopez Letona

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## Education

### Carnegie Mellon University

*Bachelor of Science in Computer Science*

Pittsburgh, PA

*Expected May 2027*

- QPA: 3.54/4.00
- Concentration in Software Engineering & Minor in Social and Political History
- Activities: Acts2Fellowship, Spanish and Latin Student Association, Grappling Club

### Bloomfield High School

*High School Diploma*

Bloomfield, NJ

*June 2023*

- GPA: 4.665
- Activities: Peer Leadership, Varsity Wrestling Captain, Honor Societies (NHS, Math, Science, Language), Red Cross Club, Glamour Gals Club, Varsity Soccer

## Experience

### CS and Mobile Development Instructor

*Prime Factors Learning Lab*

Newark, NJ

*June 2025 – August 2025*

- Designed and delivered a complete curriculum, introducing 30 students to core CS concepts, including binary systems, data structures, GitHub workflows, and mobile development with Swift and Python.
- Directed a student team in building a fullstack iOS app, integrating backend features with plans to publish to the App Store (see Swift Academy under Projects).

### CS and Mobile Development Instructor

*Prime Factors Learning Lab*

Newark, NJ

*May 2024 – August 2024*

- Taught CS and mobile dev fundamentals; guided a student team through a frontend iOS app prototype.

## Projects

### Swift Academy

[estuardolopez.com/projects/swift-academy](https://estuardolopez.com/projects/swift-academy)

PFL

*June 2025 – Present*

- Developed a fullstack iOS learning app featuring a zig-zag lesson flow and scalable MVVM structure.
- Built an AI-powered Code Coach using Python, FastAPI, and OpenAI API, with plans for App Store release.

### Hive

[estuardolopez.com/projects/hive](https://estuardolopez.com/projects/hive)

Independent

*April 2025 – May 2025*

- Fullstack Hive game with Python backend (FastAPI, Pytest, Poetry) and React (TypeScript) frontend.
- Designed a modular MVC system with clean RESTful API endpoints and scalable game state. Turn-based gameplay with a responsive hex grid, drag/zoom, action highlights, and graph-based movement logic.

### Gcode Agent

[estuardolopez.com/projects/gcode-agent](https://estuardolopez.com/projects/gcode-agent)

PFL

*March 2025 – May 2025*

- Contributed to an AI chatbot with the Mastra framework, focusing on backend design, behaviors, and RAG.
- Improved DevOps with Docker, Codespaces, and GitHub Actions for local development, testing, and CI.

### Personal Website

[estuardolopez.com/projects/personal-website](https://estuardolopez.com/projects/personal-website)

Independent

*May 2025 – June 2025*

- Responsive developer showcase with Next.js and Tailwind CSS, using interactive components and Vercel.

### Santorini

[estuardolopez.com/projects/santorini](https://estuardolopez.com/projects/santorini)

CMU

*March 2025 – April 2025*

- Fullstack Santorini game with Java backend (NanoHTTPD), includes extensive UML documentation.

## Skills

- **Programming Languages:** Python, TypeScript, Java, C, JavaScript, HTML/CSS, Swift, Standard ML
- **Technologies:** FastAPI, RESTful APIs, Next.js, React, SwiftUI, Node.js, LLM APIs, Tailwind CSS, Docker
- **Tools & Dev:** GitHub, CI/CD, Testing, DevOps, UI/UX, Package Managers, OOP, Functional Programming, Data Structures and Algorithms
- **Soft Skills:** Communication, Collaboration, Adaptability, Organization, Leadership